



The effect of the increased surface area

of one side of a 3-inch x 5-inch note card.

When the 1 cm^3 is filled with micrometre-sized cubes – a trillion (10^{12}) of them, each with a surface area of $6 \mu\text{m}^2$ – the total surface area amounts to 6 m^2 , or about the area of the main bathroom in an average house.

And when that single cubic centimetre of volume is filled with 1 nm -sized cubes – 10^{21} of them, each with an area of 6 nm^2 – their total surface area comes to $6,000 \text{ m}^2$. In other words, a single cubic centimetre of cubic nanoparticles has a total surface area larger than a football field!

Large surface area translates to:

1. Better reactivity
2. Creation of better catalysts
3. Use of nanostructured membranes and materials for water treatment and desalination
4. Support to “functionalisation” of nanoscale material surfaces.

As of 2003, catalyst technologies accounted for over \$1 trillion of revenue in the US economy and about a third of the material GDP. This lets you gauge the potential that Nanotechnology can help us unlock. Nanotechnology is not simply working at ever smaller dimensions; rather, working at the nanoscale enables scientists to utilise the unique physical, chemical, mechanical and optical properties of materials that naturally occur at that scale.

Do we need Nanotechnology?

We need nanotechnology. There are no ifs and buts about it. It will raise our standard of living, make our lives more secure, improve healthcare delivery and optimise our use of limited resources. We can already see many of the possibilities. New products that solve new problems in new ways are more difficult to foresee, yet their impact is likely to be even greater.